A $10 Chip That Could Put a Medical Lab on Your Wrist

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Imagine checking your blood sugar without a needle, using a sensor on your wrist. Researchers at the University of Cambridge and a start-up called GlitterinTech have built a light-reading chip that could one day make that possible. It costs just $10.

The chip is an optical **spectrometer**, a tool that studies light to figure out what a substance is made of. **Spectrometers** are essential in medicine, factories, food testing and environmental research, but shrinking them usually means giving up accuracy. This one is only a few centimeters across, yet the team says its **precision** rivals much bigger laboratory machines.

The secret is mathematics. The device, called a convolutional **spectrometer**, analyzes light using a rule called the convolution theorem, performing the math directly on the light itself instead of relying on heavy computer calculations afterward. It works across a range of **near-infrared** light and finishes a measurement in less than a second.

In tests, the chip correctly identified plastics, medicines, coffee, flour and tea every time. It also measured human health markers, including skin moisture, blood alcohol, blood lactate and blood **glucose**, under conditions designed to mimic the body. In one demonstration, it tracked one person's blood **glucose** without needles over an extended period. The researchers caution that much more testing is needed before it could become a real medical product.

'The key insight was to go back to the mathematics and ask whether there was a fundamentally cleaner way to retrieve **spectra**,' said Dr. Chunhui Yao, a lead author of the study.

The chip also kept working from -20°C to 80°C, a range that could suit wearables, factories and outdoor sensors. Professor Qixiang Cheng, who led the project, said what stands out is that the packaged system works reliably in real-world conditions, not just in the lab.

Word Treasure

spectrometer -- A tool that studies light to figure out what a substance is made of.
near-infrared -- Light just beyond the red light our eyes can see.
spectra -- Patterns of light that can reveal what something is made of.
glucose -- A sugar found in the blood, also called blood sugar.
precision -- How exact and accurate a measurement is.

Background you need

A spectrometer studies light that passes through or bounces off a substance to work out what that substance is made of.

Doctors and scientists use such tools to check health markers, food, medicines and materials, but large laboratory machines are hard to carry around.

Making a spectrometer small, cheap and accurate could let sensors be worn on the body or used in factories and outdoors.

Break it down

LEAD: Cambridge researchers built a $10 light-reading chip that could one day check blood glucose without needles

KEY EVENT: The tiny convolutional spectrometer matched big lab machines in tests

+- WHO: University of Cambridge researchers and the start-up GlitterinTech

+- HOW: It performs mathematics directly on near-infrared light

EVIDENCE: It identified plastics, medicines, coffee, flour and tea every time and measured health markers including blood glucose

OPEN QUESTION: How much more testing is needed before it can become a real medical product

Why it matters

A cheap, accurate chip could move health checks and material testing out of big laboratories and into wearables, factories and outdoor sensors. That could make useful measurements faster and available in many more places.

Test what you remember

1. How much does the Cambridge chip cost?

- (A) $1
- (B) About $1,000
- (C) $100
- (D) $10

2. What does a spectrometer do?

- (A) It studies light to figure out what a substance is made of
- (B) It counts blood cells
- (C) It takes photographs of the sky
- (D) It measures temperature only

3. What is the chip's device called?

- (A) A light sensor
- (B) A wrist spectrometer watch
- (C) A convolutional spectrometer
- (D) A glucose meter

4. Which materials did the chip identify correctly every time in tests?

- (A) Plastics, medicines, coffee, flour and tea
- (B) Wood, glass and metal
- (C) Sand, soil and water
- (D) Only coffee and tea

5. What do the researchers caution about the chip as a medical product?

- (A) Much more testing is needed before it could become a real medical product
- (B) It cannot measure glucose
- (C) It is too expensive to make
- (D) It only works in the dark

6. Across what temperature range did the chip keep working?

- (A) From -80°C to 20°C
- (B) From 0°C to 20°C
- (C) Only at room temperature
- (D) From -20°C to 80°C

Answer key (try the quiz first!): 1: D 2: A 3: C 4: A 5: A 6: D

Questions to think about

1. Dr. Chunhui Yao: A lead author of the study, he said the key insight was to go back to the mathematics and ask whether there was a fundamentally cleaner way to retrieve spectra.

2. Professor Qixiang Cheng: He led the project and said what stands out is that the packaged system works reliably in real-world conditions, not just in the lab.

3. The researchers: They caution that much more testing is needed before the chip could become a real medical product.

MY THOUGHTS (at least 20 words):